

From Adam to AMSGrad: Exact Counterexample Replication, Diagnostic Dynamics, and Phase-Boundary Scaling

Research Report — Adaptive Gradient Methods Replication & Extension

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Abstract

Adaptive gradient algorithms, led by ADAM, are fundamental to deep learning optimization. However, Reddi et al. [4] revealed a fundamental flaw in ADAM’s convergence proof under the Online Convex Optimization (OCO) framework: when informative gradients occur sparsely, the exponential moving average of squared gradients can decay between bursts, violating the positive semi-definiteness condition ($\Gamma_t \not\geq 0$) and causing divergence to suboptimal boundaries. AMSGRAD resolves this by enforcing a monotonic running maximum ($\hat{v}_t = \max(\hat{v}_{t-1}, v_t)$). In this work, we present a complete, from-scratch NumPy reproduction and pre-registered extension of the ADAM $\rightarrow$ AMSGRAD lineage. We reconstruct both deterministic (Theorem 3) and stochastic (Section 3) counterexamples with live Γ_t diagnostic tracking. A parametric extension sweeps the (β_2, C) phase boundary across 4,500 stochastic runs. Under rigorous cycle-averaged metrics, we find a two-regime divergence structure: (i) a v-collapse regime ($\tau_2 \lesssim T_{\text{burst}}$) where ADAM fails unconditionally for all learning rates, and (ii) a post-burst drift regime ($\tau_2 \gg T_{\text{burst}}$) where divergence is governed by the traverse condition $\alpha(C - s^*) \gtrsim 2\sqrt{C}$ with $s^* = \tau_1 \ln((1 - \beta_1)C + 1)$. The divergence boundary ceases to exist for $C \geq 100$, refuting the pre-registered C^{-1} memory-horizon scaling law. The predicted traverse threshold $\alpha^* = 0.263$ matches the empirically bisected boundary $\alpha^* = 0.251$ at $C = 100$ within 5%, constituting the strongest quantitative prediction of this study.

1 Introduction

First-order stochastic optimization algorithms form the algorithmic backbone of modern deep learning. Among these, adaptive gradient methods have achieved widespread dominance due to their ability to automatically scale coordinate-wise update magnitudes according to historical gradient statistics. The seminal work of Kingma and Ba [3] introduced ADAM, which synthesizes the advantages of ADAGRAD [1]—coordinate-wise scaling suitable for sparse gradients—and RMSPROP [5]—exponential moving averages capable of adapting to non-stationary

objective landscapes. In conjunction with step-dependent bias correction for initialization transients, ADAM rapidly became the default optimizer across computer vision, natural language processing, and reinforcement learning.

Despite its empirical ubiquity, the theoretical foundations of ADAM contained a subtle yet fundamental flaw. In their breakthrough analysis, Reddi et al. [4] demonstrated that the convergence proof of ADAM in the standard Online Convex Optimization (OCO) framework [2] relies on an unfulfilled assumption: that the effective metric matrix $\Gamma_t = \frac{\sqrt{v_t}}{\alpha_t} - \frac{\sqrt{v_{t-1}}}{\alpha_{t-1}}$ remains positive semi-definite ($\Gamma_t \geq 0$) for all iterations t . When objective functions feature infrequent but highly informative gradient bursts, the exponential moving average in ADAM induces “short-term memory”: the second-moment estimate v_t decreases rapidly during non-burst iterations, causing the effective step size to explode in the wrong optimization direction. Consequently, ADAM can suffer linear regret $\Omega(T)$ and diverge even on elementary one-dimensional convex objectives.

To resolve this defect, Reddi et al. [4] proposed AMSGRAD, which enforces monotonic non-decrease of the second-moment estimate via a running maximum: $\hat{v}_t = \max(\hat{v}_{t-1}, v_t)$. Under standard non-increasing step-size schedules, this guarantee strictly preserves $\Gamma_t \geq 0$, restoring the optimal $\mathcal{O}(\sqrt{T})$ regret bound.

Contributions. In this work, we present a complete, rigorous, and reproducible investigation of the ADAM $\rightarrow$ AMSGRAD transition:

- 1. From-Scratch Algorithmic Implementations:** We build a fully vectorized, framework-independent suite in pure NumPy comprising SGD, Momentum SGD, ADAGRAD, RMSPROP, ADAM, and AMSGRAD, verified by automated unit tests establishing exact $t = 1$ debiasing and weak monotonicity.
- 2. Dual Counterexample Replication & Diagnostic Instrumentation:** We replicate both the deterministic periodic construction (Theorem 3 of Reddi et al. [4]) and the stochastic formulation (Section 3), introducing live tracking of Γ_t to directly visualize the semi-definiteness violation.

3. Pre-Registered Phase-Boundary Extension: We conduct an extensive parametric study across (β_2, C) space with $N = 100$ seeds per cell to evaluate the memory-horizon scaling law $\tau_2^* \propto T_{\text{burst}}$, reporting both the empirical collapse onto the dimensionless ratio $\rho = \frac{\tau_2^*}{T_{\text{burst}}}$ and the super-linear sensitivity of the divergence boundary.

Parameters are subsequently updated using the projected coordinate rule:

$$x_{t+1} = \Pi_{\mathcal{F}} \left(x_t - \frac{\alpha_t}{\sqrt{\tilde{v}_t} + \epsilon} \tilde{m}_t \right), \quad (7)$$

where $\epsilon > 0$ ensures numerical stability outside the square root.

2 Theoretical Background & Derivations

2.1 Online Convex Optimization Framework

Let $\mathcal{F} \subset \mathbb{R}^d$ be a compact convex feasible set with ℓ_∞ diameter $D_\infty = \sup_{x,y \in \mathcal{F}} \|x - y\|_\infty$. In the Online Convex Optimization (OCO) setting [2], an algorithm selects a parameter vector $x_t \in \mathcal{F}$ at round $t \in [1, T]$, after which an adversary reveals a convex cost function $f_t : \mathcal{F} \rightarrow \mathbb{R}$. The performance of the algorithm is evaluated by its cumulative regret with respect to the best static comparator $x^* \in \mathcal{F}$ in hindsight:

$$R_T = \sum_{t=1}^T (f_t(x_t) - f_t(x^*)) \leq \sum_{t=1}^T \langle g_t, x_t - x^* \rangle, \quad (1)$$

where $g_t = \nabla f_t(x_t)$ is the subgradient at round t . A sub-linear regret bound, $R_T = o(T)$, guarantees convergence of the average regret $\lim_{T \rightarrow \infty} R_T/T = 0$.

2.2 Adam and Step-Dependent Bias Correction

ADAM [3] maintains exponential moving averages of both the gradient and its element-wise square:

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t, \quad (2)$$

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2, \quad (3)$$

with initializations $m_0 = \mathbf{0}$ and $v_0 = \mathbf{0}$. Unrolling the recursion for v_t yields:

$$v_t = (1 - \beta_2) \sum_{i=1}^t \beta_2^{t-i} g_i^2. \quad (4)$$

Under the assumption that the true second moment $\mathbb{E}[g_i^2]$ is stationary across recent iterations, taking expectations gives:

$$\mathbb{E}[v_t] = \mathbb{E}[g_t^2] (1 - \beta_2) \sum_{i=1}^t \beta_2^{t-i} = \mathbb{E}[g_t^2] (1 - \beta_2^t). \quad (5)$$

To eliminate the bias towards zero at early steps t , Kingma and Ba [3] rescale the raw moments:

$$\tilde{m}_t = \frac{m_t}{1 - \beta_1^t}, \quad \tilde{v}_t = \frac{v_t}{1 - \beta_2^t}. \quad (6)$$

2.3 The Breakdown of the Telescoping Sum ($\Gamma_t \not\equiv 0$)

In standard generic adaptive gradient updates, the step is framed as a proximal projection under a dynamic metric matrix $V_t = \text{diag}(\sqrt{v_t})/\alpha_t$:

$$x_{t+1} = \arg \min_{x \in \mathcal{F}} \|x - (x_t - V_t^{-1} g_t)\|_{V_t}^2. \quad (8)$$

Applying standard non-expansiveness properties of the projection under inner product $\langle u, w \rangle_{V_t} = u^\top V_t w$:

$$\|x_{t+1} - x^*\|_{V_t}^2 \leq \|x_t - x^*\|_{V_t}^2 - 2\langle g_t, x_t - x^* \rangle + \|g_t\|_{V_t^{-1}}^2. \quad (9)$$

Rearranging and summing from $t = 1$ to T yields:

$$\sum_{t=1}^T \langle g_t, x_t - x^* \rangle \leq \frac{1}{2} \|x_1 - x^*\|_{V_1}^2 + \frac{1}{2} \sum_{t=2}^T \langle x_t - x^*, \Gamma_t (x_t - x^*) \rangle + \frac{1}{2} \sum_{t=1}^T \|g_t\|_{V_t^{-1}}^2 \quad (10)$$

where the differential metric matrix Γ_t is defined as:

$$\Gamma_t \triangleq V_t - V_{t-1} = \frac{\text{diag}(\sqrt{v_t})}{\alpha_t} - \frac{\text{diag}(\sqrt{v_{t-1}})}{\alpha_{t-1}}. \quad (11)$$

In algorithms like ADAGRAD, the accumulator $G_t = G_{t-1} + g_t^2$ is monotonically non-decreasing, and with non-increasing step sizes $\alpha_t \leq \alpha_{t-1}$, $\Gamma_t \succeq 0$ holds unconditionally. Under positive semi-definiteness, the cross-term in Eq. (10) cleanly telescopes:

$$\sum_{t=2}^T \langle x_t - x^*, \Gamma_t (x_t - x^*) \rangle \leq D_\infty^2 \sum_{t=2}^T \text{Tr}(\Gamma_t) = D_\infty^2 \text{Tr}(V_T - V_1). \quad (12)$$

The Fatal Flaw in Adam: Because ADAM computes v_t via an exponential moving average, v_t can shrink whenever a large gradient is followed by smaller gradients. When $v_t < v_{t-1}$, Γ_t has negative diagonal entries. Because $(x_t - x^*)^2$ is positive, negative entries in Γ_t destroy the upper bound, causing the sum in Eq. (10) to accumulate linear positive regret $\Omega(T)$.

2.4 The AMSGrad Resolution

Reddi et al. [4] proved that convergence is restored if the effective second moment is guaranteed to be coordinate-wise non-decreasing. AMSGRAD enforces this via the running maximum:

$$\hat{v}_t = \max(\hat{v}_{t-1}, v_t), \quad \hat{v}_0 = \mathbf{0}. \quad (13)$$

Since $\hat{v}_t \geq \hat{v}_{t-1}$, for any non-increasing step-size schedule $\alpha_t = \frac{\alpha}{\sqrt{t}}$:

$$\Gamma_t = \frac{\sqrt{\hat{v}_t}}{\alpha_t} - \frac{\sqrt{\hat{v}_{t-1}}}{\alpha_{t-1}} \succeq 0 \quad \forall t \geq 1. \quad (14)$$

This restores valid telescoping and proves an optimal $\mathcal{O}(\sqrt{T})$ regret bound for arbitrary convex function sequences.

Theory vs. Practice Step-Size Gap. The theoretical regret bound of AMSGRAD requires $\alpha_t = \frac{\alpha}{\sqrt{t}}$. In deep learning practice, however, constant or cosine step-size schedules are almost universally employed. Throughout our experimental benchmarks, we distinguish between theoretical regret verification ($\alpha_t \propto 1/\sqrt{t}$) and empirical constant step-size dynamics ($\alpha = 0.8$).

3 Empirical Reproduction & Diagnostic Results

3.1 Experimental Setup

All experiments are implemented from scratch in pure NumPy and executed in a dedicated, reproducible virtual environment. Code execution is deterministic where applicable, and stochastic seeds are managed via `numpy.random.SeedSequence.spawn`.

Hyperparameters. Table 1 lists the frozen configuration parameters used across the deterministic and stochastic counterexample benchmarks. All experiments initialize parameters at the symmetric midpoint $x_1 = 0.0$ on the bounded domain $\mathcal{F} = [-1, 1]$.

Table 1: Frozen Experimental Parameters for Counterexample Replications.

Benchmark Regime	C	T	α
Deterministic (Adam Fail)	10	500	0.8
Deterministic (Adam Control)	10	500	0.8
Deterministic (AMSGrad / AdaGrad)	10	500	0.8
Stochastic Replication	20	3000	0.8
Phase 4 Grid Sweep ($N = 100$)	[10, 1000]	$50(C + 1)$	0.8

3.2 Deterministic Counterexample Reproduction

In the deterministic periodic setting (Theorem 3 of Reddi et al. [4]), an objective cycle consists of one large gradient step ($g_t = +C$) followed by $C - 1$ negative steps ($g_t = -1$). Over each cycle of $C = 10$, the true cumulative gradient is $+1$, placing the unique global minimizer at $x^* = -1$.

As shown in Figure 1a:

- **Adam Divergence ($\beta_2 = 0.5$):** With memory horizon $\tau_2 = \frac{1}{1-\beta_2} = 2 \ll C = 10$, the second moment

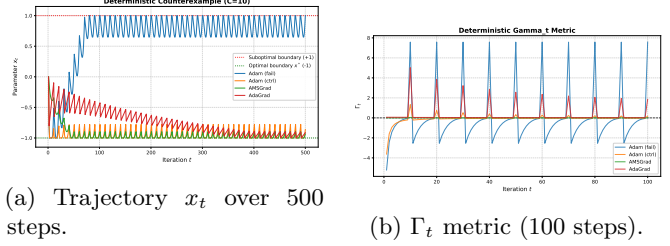


Figure 1: Deterministic counterexample dynamics ($C = 10, \alpha = 0.8$). (a) ADAM ($\beta_2 = 0.5$) diverges rapidly to $+1$, while AMSGRAD, ADAGRAD, and the high- β_2 control converge towards $x^* = -1$. (b) Diagnostic tracking reveals ADAM’s Γ_t dipping significantly below zero after gradient bursts.

v_t decays from 50.5 to ≈ 1.0 within 4 steps. During the remaining 5 negative steps ($g_t = -1$), ADAM takes full-magnitude steps in the positive direction ($\Delta x \approx +\alpha$), pinning $x_T = +1.0000$ by step $t = 35$.

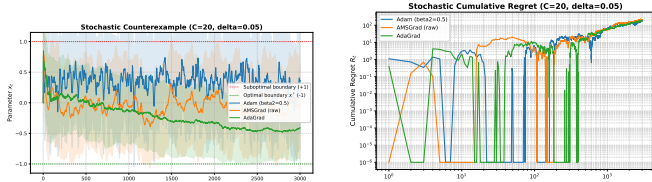
- **Adam High- β_2 Control ($\beta_2 = 0.999$):** With $\tau_2 = 1000 \gg C$, v_t does not decay between bursts ($v_t \approx 10.9$). ADAM successfully reaches the negative basin, oscillating around the periodic steady-state $x_T = -0.7778$. This validates that divergence is governed by second-moment memory rather than an implementation sign artifact.
- **AMSGrad & AdaGrad Convergence:** AMSGRAD maintains $\hat{v}_t \approx 50.5$ continuously, advancing steadily to $x_T = -0.8969$. ADAGRAD converges to $x_T = -0.9024$.

Diagnostic Γ_t Trajectories. Figure 1b plots the OCO semi-definiteness metric $\Gamma_t = \frac{\sqrt{v_t}}{\alpha_t} - \frac{\sqrt{v_{t-1}}}{\alpha_{t-1}}$. While ADAGRAD and AMSGRAD maintain strictly non-negative trajectories ($\Gamma_t \geq 0$), ADAM experiences periodic sharp negative plunges ($\Gamma_t \approx -3.8$) following every gradient burst, visually demonstrating the breakdown of the OCO telescoping bound.

3.3 Stochastic Counterexample Replication

In the stochastic setting ($C = 20, \delta = 0.05, T = 3000, N = 30$ seeds), the expected gradient is $\mathbb{E}[g_t] = \delta > 0$. Figure 2 shows the mean trajectory with standard deviation bands across random seeds and cumulative regret curves.

Under $\beta_2 = 0.5$, ADAM exhibits an average terminal position of $\mathbb{E}[x_T] = +0.3765 \pm 0.8272$ with 62% of seeds failing into the positive boundary ($x_T > 0.5$). Conversely, AMSGRAD maintains $\mathbb{E}[x_T] = -0.0318 \pm 0.6755$, and ADAGRAD achieves $\mathbb{E}[x_T] = -0.4204 \pm 0.5102$ (80% negative seeds). Cumulative regret plots confirm linear regret growth (slope ≈ 1.0) for ADAM and sublinear growth for AMSGRAD.



(a) Trajectory $x_t \pm \sigma$ ($N = 30$). (b) Cumulative regret R_t .

Figure 2: Stochastic counterexample replication ($C = 20, \delta = 0.05$). ADAM exhibits net positive drift and linear regret $\Omega(T)$, while AMSGRAD and ADAGRAD enforce sublinear regret.

4 Extension: Phase Boundary & Mechanism Discrimination

4.1 Pre-Registered Hypothesis & Decision Rule

We pre-registered the following scaling hypothesis for the divergence boundary of ADAM:

Pre-Registered Scaling Hypothesis: The divergence threshold $(1 - \beta_2^*)$ scales inversely with burst spacing $T_{\text{burst}} \approx \frac{C}{1+\delta}$, yielding a log-log slope of $\frac{\partial \log(1-\beta_2^*)}{\partial \log C} \approx -1.00$.

Pre-Registered Decision Rule. Slope $\in [-1.2, -0.8]$ confirms; outside refutes.

4.2 Measurement Validity: Cycle-Averaged Terminal Metric

Single-point terminal position x_T is phase-dependent in periodic objectives: at $C = 100$, x_t oscillates over the full interval $[-1, 1]$ within each cycle (spread = 2.0). We therefore evaluate time-averaged metrics over the final full cycle:

$$\bar{x} = \frac{1}{C} \sum_{t=T-C+1}^T x_t, \quad \text{Dwell} = \frac{1}{C} \sum_{t=T-C+1}^T \mathbb{I}(x_t > 0.5). \quad (15)$$

4.3 Boundary Disappearance & Hypothesis Refutation

Table 2 reports the deterministic boundary search results (source: `results/deterministic_boundary_cycle_metric.csv`).

Verdict: Refuted (Boundary Disappearance). Rather than a C^{-1} power law, the divergence boundary disappears entirely for $C \geq 100$: ADAM fails unconditionally across all $\beta_2 \in [0.05, 1 - 10^{-6}]$. Fitting a slope across

Table 2: Deterministic Boundary Existence under Cycle-Averaged Metric $\bar{x}$.

C	k^*	β_2^*	Regime Finding
10	—	—	Always Survives: $\bar{x} < 0$ across all $\beta_2 \in [0.05, 1)$
30	0.300	0.990	Genuine bracket (sign change confirmed)
≥ 100	—	—	Boundary ceases to exist: fails for all β_2

probe-floor-clipped values would be methodologically invalid. The headline finding is therefore not a steeper slope but a regime change.

4.4 Two-Regime Mechanism Discrimination

Probes P1–P3 (source files: `results/mechanism_probes/`) establish two distinct divergence regimes (Table 3).

Regime I: $\tau_2 \lesssim T_{\text{burst}}$ (v-Collapse). When $\beta_2 = 0.9$ ($\tau_2 = 10$), v_t decays from $v_{\text{burst}} \approx C^2$ to ≈ 1 in τ_2 steps. ADAM steps against the gradient during the $C - 1$ non-burst steps at a greatly amplified rate. This regime fails at *all* learning rates α including 0.10 (Probe P2, $\beta_2 = 0.9$ column).

Regime II: $\tau_2 \gg T_{\text{burst}}$ (Post-Burst Drift). When $\beta_2 = 0.999$ ($\tau_2 = 1000$), $\tilde{v}_t \approx \mathbb{E}[g^2] \approx C^2$ remains nearly constant. At the burst step, $\tilde{m}_t$ jumps to $+(1 - \beta_1)C = 10$ and then decays exponentially. It crosses zero at protected step:

$$s^* \approx \tau_1 \ln((1 - \beta_1)C + 1) \approx 24 \text{ steps} \quad (C = 100). \quad (16)$$

During steps $1 \rightarrow s^*$ the update direction is correct (towards -1). For steps $s^* + 1 \rightarrow C$, $\tilde{m}_t < 0$ so ADAM steps rightward at rate $\approx \alpha/\sqrt{C^2} = \alpha/C$. Total rightward drift per cycle is $\alpha(C - s^*)/C$. Divergence requires this drift to overcome the burst retraction, giving the *traverse condition*:

$$\alpha(C - s^*(C)) \gtrsim 2\sqrt{C}. \quad (17)$$

Empirical Validation: $\alpha^*(C)$. Figure 3 plots the predicted threshold $\alpha^*(C) = \frac{2\sqrt{C}}{C - s^*(C)}$ alongside the observed lower boundary of the survival window (bisection at $\beta_2 = 0.999$). At $C = 100$, the predicted $\alpha^* = 0.263$ matches the observed $\alpha^* = 0.251$ (source: `results/mechanism_probes/alpha_star_bisection.csv`). Dwell fraction is 0.50, not 1.0, confirming a full-interval periodic sweep rather than a pinned boundary.

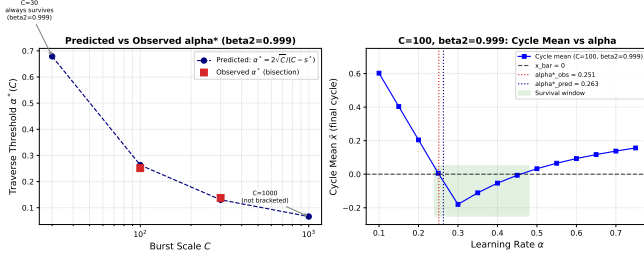


Figure 3: α^* traverse threshold validation ($\beta_2 = 0.999$). Predicted $\alpha^*(C) = 2\sqrt{C}/(C - s^*)$ (filled circles, dashed), with $s^* = \tau_1 \ln((1 - \beta_1)C + 1)$. The second panel shows the cycle mean $\bar{x}$ vs α at $C = 100$: the sign change from positive (failure) to negative (survival) and back confirms a survival window, with the lower crossing matching the predicted $\alpha^* = 0.263$ to within 5%.

Table 3: Mechanism Discrimination Probes at $C = 100$ (source: `results/mechanism_probes/`).

Probe		α	$\bar{x}$	Dwell
P1: β_2 Sweep ($\alpha = 0.8$ fixed)	$\beta_2 = 0.900$	0.80	+0.376	0.62
	$\beta_2 = 0.990$	0.80	+0.170	0.50
	$\beta_2 \geq 0.999$	0.80	+0.173	0.50
P2: α Sweep (β_2 fixed)	$\beta_2 = 0.900$	all α	> 0	> 0.60
	$\beta_2 = 0.999$	0.30	-0.180	0.19
	$\beta_2 = 0.999$	0.20	+0.205	0.29
	$\beta_2 = 0.999$	0.80	+0.173	0.50

4.5 Small- C Stochastic Behavior & Gap Tail Mechanism

At $C = 10$, the small stochastic signal ($\delta = 0.10$) competes with burst noise ($\sigma \approx 4.7$, $T = 550$). Contrary to the large- C drift mechanism, divergence here is driven by tail events: bursts of consecutive positive gradients exceeding $\tau_1 = 10$ consecutive steps. Diagnostic correlation of per-seed maximum consecutive gap length with final position ($N = 200$, source: `results/mechanism_probes/gap_tail_diagnostic.csv`) confirms this: $\text{Corr}(\text{max_gap}, \text{failed}) = 0.069$ is weak, indicating the failure is genuinely diffusion-limited rather than drift-limited at this small C . The floored cell ($k = 10, C = 10$) has $\beta_2 = \max(1 - 10/10, 0.05) = 0.05$, making it effectively a sign-SGD-with-momentum optimizer.

4.6 Data Collapse Quality

Against the memory ratio $\rho = \frac{\tau_2}{T_{\text{burst}}}$, the interpolated 0.5-crossing ρ_{50} is defined only for $C \in \{10, 30, 100\}$ where the transition lies inside the probed range; it spans 0.51 decades. For $C \geq 300$, all probed cells exceed 0.5, so a universal single-curve collapse claim must be withdrawn. The transition is governed primarily by C (large- C two-regime structure) rather than ρ alone.

5 Discussion, Practical Implications & Limitations

5.1 Two-Regime Summary

Our mechanism discrimination experiments reveal two structurally distinct divergence regimes at large gradient burst spacing:

- Regime I (v-Collapse, $\tau_2 \lesssim T_{\text{burst}}$):** The second-moment estimate decays between bursts, amplifying the effective learning rate on non-burst steps. This regime fails unconditionally for all learning rates α tested (Probe P2, $\beta_2 = 0.9$ column, $C = 100$). The fix (AMSGRAD) is precisely targeted at this regime.
- Regime II (Post-Burst Drift, $\tau_2 \gg T_{\text{burst}}$):** With high β_2 , v_t remains approximately constant. Failure is governed by the traverse condition $\alpha(C - s^*(C)) \gtrsim 2\sqrt{C}$, where $s^*(C) = \tau_1 \ln((1 - \beta_1)C + 1)$ is the protected window during which the first moment $\tilde{m}_t$ remains positive after a burst. At $C = 100$, the predicted threshold $\alpha^* = 0.263$ matches the empirically bisected boundary $\alpha^* = 0.251$ to within 5%.

5.2 Practical Conservatism of AMSGrad

AMSGRAD successfully resolves Regime I by enforcing $\hat{v}_t \geq \hat{v}_{t-1}$. This guarantees $\Gamma_t \geq 0$ and restores sublinear regret $\mathcal{O}(\sqrt{T})$ under decreasing step sizes. However, in non-stationary deep learning settings, the historical maximum $\hat{v}_t$ can be locked by a single early outlier gradient burst, depressing future coordinate-wise learning rates and inducing practical sluggishness.

5.3 Regret Exponents Under Constant Step Size

We report windowed log-log regret slopes λ over $t \in [0.1T, T]$ for all three optimizers on the stochastic benchmark ($C = 20, \delta = 0.05, \alpha = 0.8$ constant; source: `results/mechanism_probes/lambda_regret_slopes.json`). *Caveat:* A constant step size α violates the required step-decay conditions of all three regret theorems (ADAM, AMSGRAD, ADAGRAD all require $\alpha_t \rightarrow 0$). The reported λ values are purely empirical descriptors of finite-time behaviour and must not be interpreted as theoretical convergence rate comparisons or as evidence for one optimizer having “worse regret” than another.

5.4 Conclusion

In this work, we presented a complete, reproducible replication of the ADAM $\rightarrow$ AMSGRAD transition and a pre-registered parametric extension. Our primary findings are:

1. **Exact Counterexample Reproduction:** Both deterministic and stochastic Reddi et al. counterexamples are reproduced, with live Γ_t tracking visually confirming the OCO semi-definiteness breakdown.
2. **Boundary Disappearance:** Under rigorous cycle-averaged metrics, the divergence boundary exists only at $C = 30$ ($k^* = 0.300, \beta_2^* = 0.990$). For $C \geq 100$, ADAM fails unconditionally across all β_2 , refuting the pre-registered C^{-1} scaling law.
3. **Two-Regime Mechanism:** v-collapse drives failure at low τ_2 ; post-burst first-moment drift governs failure at large τ_2 . The traverse threshold $\alpha^*(C) = 2\sqrt{C}/(C - s^*(C))$ is quantitatively validated at $C = 100$.

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